

A1.

- (a) $A \cup B = \{3, 4, 5, 6, 7\}$, $|A \cup B| = 5$
(b) $A \cap B = \{5, 6\}$, $|A \cap B| = 2$
(c) $(A \cap B) \cup C = \{1, 5, 6, \{3, 5\}, \{3, 7\}\}$, $|(A \cap B) \cup C| = 5$
(d) $P(A) \cap C = \{\{3, 5\}, \{3, 7\}\}$, $|P(A) \cap C| = 2$
(e) $(A \cap B) \times C = \{(5, 1), (5, \{3, 5\}), (5, \{3, 7\}), (6, 1), (6, \{3, 5\}), (6, \{3, 7\})\}$, $|(A \cap B) \times C| = 7$

A2.

- (a) $\{n \in \mathbb{N} : n \text{ is a prime number and } n < 10\} = \{2, 3, 5, 7\}$, set contains 4 elements
(b) $\{n \in \mathbb{Z} : n^2 < 6\} = \{-2, -1, 0, 1, 2\}$, set contains 5 elements
(c) $\{x \in \mathbb{R} : 1 < x < 2 \text{ and } x^2 \in \mathbb{Z}\} = \{\sqrt{2}, \sqrt{3}\}$, set contains 2 elements
(d) $\{x \in \mathbb{R} : |x| < 2 \text{ and } x^2 > 4\} = \emptyset$, set contains 0 elements
(e) $\{n^2 : n \in \mathbb{Z} \text{ and } 2 \leq n \leq 3\} = \emptyset$, set contains 0 elements
(f) $\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 < 4\} = \{(1, 1), (1, -1), (-1, 1), (-1, -1)\}$, set contains 4 elements
(g) $\{x \in \mathbb{R} : 1 < x < 2 \text{ and } x^2 \in \mathbb{Z}\} = \emptyset$, set contains 0 elements
(h) $A = \{6, 7, 8\} : |A| = 3 = \{ \{6, 7\}, \{6, 8\}, \{7, 8\} \}$, set contains 3 elements

A3.

Let $A = \{m+n : m, n \in \mathbb{Z} \text{ with } m \text{ even, and } n \text{ odd}\}$,
 $B = \{k \in \mathbb{Z} : k \text{ is odd}\}$

Prove that $A = B$

Proof.

$A = B \Rightarrow A \subseteq B$ and $B \subseteq A$ (equality of sets)

Let $x \in A : x = m+n$ where m is even and n is odd

Thus, let $m = 2p$ and $n = 2q+1$ where $p, q \in \mathbb{Z}$

Hence, $x = m+n$

$$= (2p) + (2q+1) = 2(p+q) + 1$$

Let $p+q = r : x = 2r+1$ where $r \in \mathbb{Z}$

Therefore x is odd $\Rightarrow x \in B$

Therefore, $A \subseteq B$

Let $k \in B : k = m+n$

Consider $m = 0, n = 1$

Thus $k = m+n$

$$= (0) + (1) = 1$$

$\Rightarrow$ condition for m, n satisfied

Therefore $k \in A$ and $B \subseteq A$

Therefore $A = B$

A4.

Let $S = \{x \in \mathbb{R} : x \geq 0\}$ and $\tau = \{u^2 + v^2 : u, v \in \mathbb{R}\}$

Prove that $S \subseteq \tau$. Is it true that $S = \tau$? [You can assume standard properties of real numbers.]

Proof.

Let $x \in S : x \geq 0$ and $x \in \mathbb{R}$

Since $x \geq 0$, then $x^2 \geq 0$

Consider $x^2 = u^2 + v^2$ where $u, v \in \mathbb{R}$

If $u = 1$ and $v = 1$, then $x^2 = u^2 + v^2$

$$= (1)^2 + (1)^2 = 2 > 0$$

Hence, $x \in \tau$ and $S \subseteq \tau$

If $(S = \tau)$ is true, then $\tau \subseteq S$ as $S \subseteq \tau$ is already proven

Let $x \in \tau : x = u^2 + v^2$ where $u, v \in \mathbb{R}$

$u^2 \geq 0 \forall u \in \mathbb{R} \Rightarrow v^2 \geq 0$ and $u^2 \geq 0$ (property of real numbers)

Since $x = v^2 + u^2$, then $x \geq 0 \Rightarrow x \in S$

Thus $\tau \subseteq S$

Therefore $S = \tau$

A5.

Let

$$U = \{k+m+n : k, m, n \in \mathbb{N} \text{ with } k, m, n \text{ all odd}\}$$

$$V = \{i \in \mathbb{N} : i \text{ is odd}\}$$

Prove that $U \subseteq V$ but $V \not\subseteq U$

Proof.

Let $x \in U : x = k+m+n$ where k, m, n are all odd

Let $k = 2p+1, m = 2q+1, n = 2r+1$ where $p, q, r \in \mathbb{N}$

(b) Then $x = k+m+n$

$$= (2p+1) + (2q+1) + (2r+1) = 2(p+q+r) + 3$$

(c) Let $s = p+q+r : x = 2s+3 \Rightarrow x \in V$

(d) Hence, $U \subseteq V$

Proof by contradiction that $V \not\subseteq U$

Assume $V \subseteq U : x \in V$ where x is odd

Let $x = k+m+n$ where $k, m, n \in \mathbb{N}$

Consider $k = 2p, m = 2q, n = 2r+1 : x = k+m+n$

$$= (2p) + (2q) + (2r+1)$$

$$= 2(p+q+r) + 1 \text{ which is odd}$$

* Satisfies condition for x in V but not in U as

for $x \in U$, k, m, n must all be odd

Thus, $|V| > |U|$ and so $V \not\subseteq U$

A6.

(a) For all x in the set of Real numbers, 2 times x is greater or equal to x = False

$$\exists x \in \mathbb{R} : 2x < x$$

(b) There exists an element n in the set of Integers such that n squared plus n is equal to 6 = True

$$\forall n \in \mathbb{Z} : n^2 + n \neq 6$$

Continue later

A7.

(a) There exists an element p in the set P such that for all s in the set S , p is the mother of s

False

(b) For all elements s in the set S , there exist an element p in the set P such that p is the mother of s

True

A8.

P	Q	$P \vee Q$	$\neg(P \vee Q)$	P	Q	$\neg P$	$\neg Q$	$(\neg P) \wedge (\neg Q)$
0	0	0	1	0	0	1	1	1
0	1	1	0	0	1	1	0	0
1	0	1	0	1	0	0	1	0
1	1	1	0	1	1	0	0	0

Hence $\neg(P \vee Q)$ is logically equivalent to $(\neg P) \wedge (\neg Q)$

Continue later and revise ' $\Rightarrow$ ', ' $\Leftarrow$ ', and ' $\Leftrightarrow$ '

B1.

(a) Consider $(A \cap B) \subseteq A \cup B$

Let $x \in (A \cap B) \Rightarrow x \in A$ and $x \in B$

But $(A \cup B) \Rightarrow x \in A$ or $x \in B$ or both

Thus satisfying $(A \cup B)$

Therefore $(A \cap B) \subseteq (A \cup B)$

∴

B2.

$$\bigcap_{n=1}^{\infty} A_n = \{x : x \in A_n \text{ for all } n \geq 1\} \text{ and } \bigcup_{n=1}^{\infty} A_n = \{x : x \in A_n \text{ for some } n \geq 1\}$$

$$\text{Let } A_n = \left\{x \in \mathbb{R} : -\frac{1}{n} < x < \frac{1}{n}\right\} \text{ for } n \in \mathbb{N}$$

$$\text{RTP } \bigcap_{n=1}^{\infty} A_n = \emptyset \text{ and find } \bigcup_{n=1}^{\infty} A_n$$